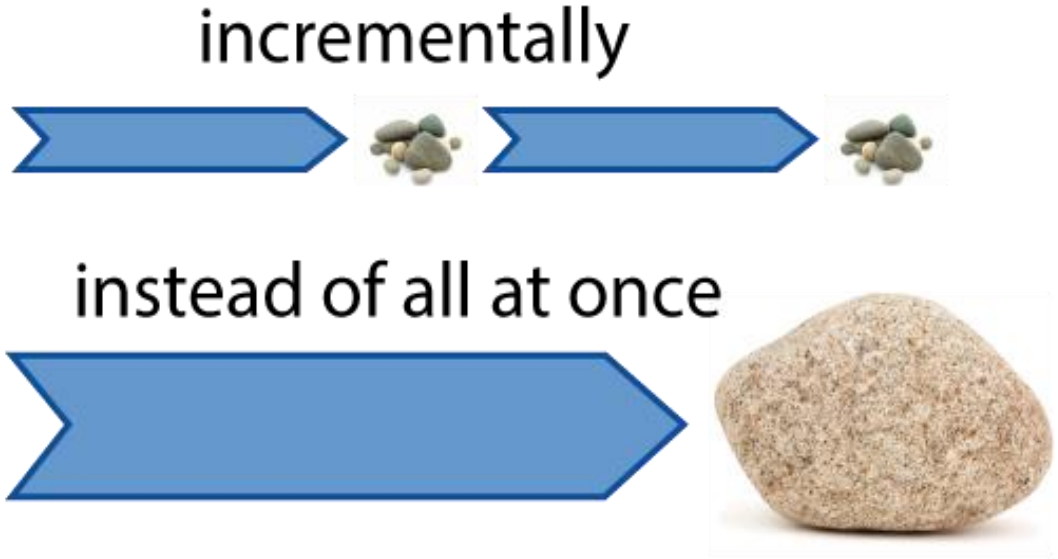
AGILE

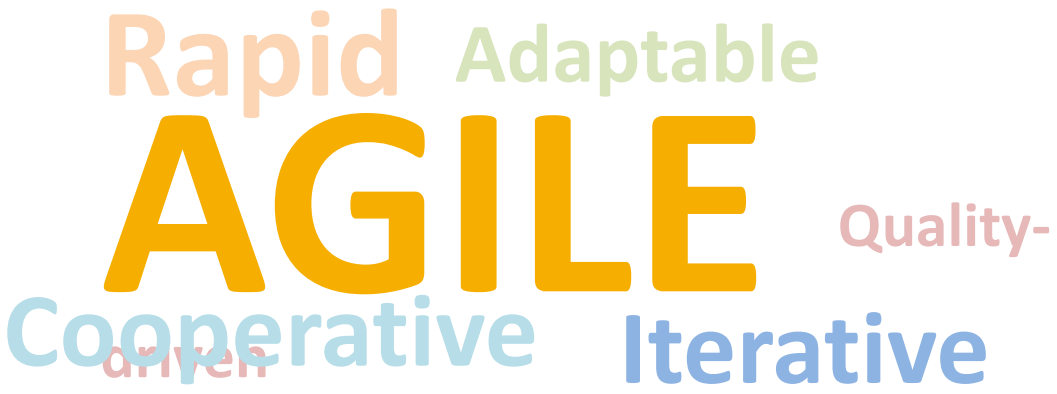
DINH THI MINH NGUYET

What is Agile?

Agile is a time boxed, iterative approach to software delivery that builds software incrementally from the start of the project, instead of trying to deliver it all at once near the end.



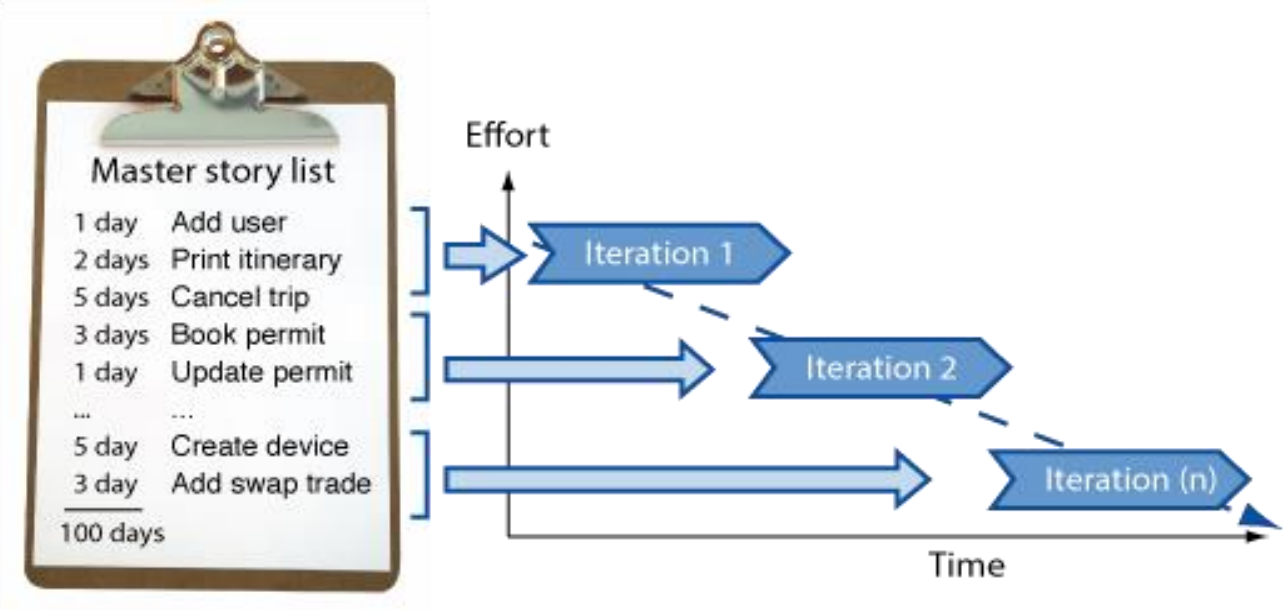
What is Agile?



Not a process, it's a **philosophy** or **set of values**

What is Agile?

It works by breaking projects down into little bits of user functionality called **user stories**, prioritizing them, and then continuously delivering them in short two week cycles called **sprints.**



Key Principles of Agile

Agile methodologies share three key principles:

1. A focus on adaptive rather than predictive methodologies
2. A focus on people rather than roles
3. A focus on self-adaptive processes

The Agile Manifesto And Mindset

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

Individuals and interactions over processes and tools

Working software over comprehensive documentation

Customer collaboration over contract negotiation

Responding to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.

© 2001, the Agile Manifesto authors

Kent Beck, Mike Beedle, Are van Bennekum, Alistair Cockburn, Ward Cunningham, Martin Fowler, James Grenning, Jim Highsmith, Andrew Hunt, Ron Jefferies, Jon Kern, Brian Marick, Robert C. Martin, Steve Mellor, Ken Schwaber, Jeff Sutherland, Dave Thomas (www.AgileAlliance.org)

The 12 Principles Behind The Agile Manifesto (1 of 2)

The highest priority is to satisfy the customer through early and continuous delivery of valuable software.

Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.

Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.

Businesspeople and developers work together daily throughout the project.

The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.

Working software is the primary measure of progress.

Kent Beck, Mike Beedle, Are van Bennekum, Alistair Cockburn, Ward Cunningham, Martin Fowler, James Grenning, Jim Highsmith, Andrew Hunt, Ron Jefferies, Jon Kern, Brian Marick, Robert C. Martin, Steve Mellor, Ken Schwaber, Jeff Sutherland, Dave Thomas (www.AgileAlliance.org)

The 12 Principles Behind The Agile Manifesto (2 of 2)

Continuous attention to technical excellence and good design enhances agility.

Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.

Simplicity—the art of maximizing the amount of work not done—is essential.

The best architectures, requirements, and designs emerge from self-organizing teams.

At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

Kent Beck, Mike Beedle, Are van Bennekum, Alistair Cockburn, Ward Cunningham, Martin Fowler, James Grenning, Jim Highsmith, Andrew Hunt, Ron Jefferies, Jon Kern, Brian Marick, Robert C. Martin, Steve Mellor, Ken Schwaber, Jeff Sutherland, Dave Thomas (www.AgileAlliance.org)

Agile Methodologies—Not for Every Project

Agile methodologies are not for everyone

An agile process is recommended if your project involves

- unpredictable or dynamic requirements
- responsible and motivated developers
- customers who understand the process and will get involved

Implementing Agile

Agile team:

- 3 to 9 members
- 100% dedicated to the teams
- Collaborative work instead of mini-waterfalls

Implementing Agile

Agile roles:

- Cross-functional team members
- Product owner
- Team facilitator

Implementing Agile

Role	Description
Cross-functional team members	Consist of team members with all the needed skills to produce a working product. Typically, a cross functional team includes designers, developers, testers, business analysts, and any other required roles.
Product owner	This role is responsible for guiding the direction of the product. Together with their team, product owners provide feedback and setting direction of functionality to be developed. The product owners rank the work based on its business value. Product owners create the backlog for and with the team.
Team facilitator	This role can be called a project manager, scrum master, project team lead, team coach, servant leader or team facilitator. Team facilitator focuses on coaching people who want to help, fostering greater collaboration on the team, align stakeholder needs, and help the team with technical project management activities.

Implementing Agile

Team workspaces:

- Create workspaces that balance common and social areas with both quiet/ private areas and open and collaborative work environments

Implementing Agile

Team workspaces:

- Create workspaces that balance common and social areas with both quiet/ private areas and open and collaborative work environments

Implementing Agile: common practices

1. Retrospective
2. Backlog preparation (only enough to understand the first release)
3. Backlog refinement
4. Daily stand-up
5. Demonstration/Reviews: at least 1 working product every 2 weeks.
6. Planning for iteration-based agile
7. Execute practices that help teams deliver value.

Agile Umbrella

* Check wikipedia for list of all Agile methods

More Prescriptive

more rules to follow

- **RUP (120+)**
  RUP has over 30 roles, over 20 activities, and over 70 artifacts
- **XP (13)**
- **Scrum (9)**
- **Kanban (3)**
- **Do Whatever!! (0)**

More Adaptive

fewer rules to follow

Scrum (1 of 4)

Originated in 1995 by Sutherland and Schwaber

Most popular methodology for agile (58%)

- Scrum framework includes
- Scrum teams with associated roles, events, artifacts, and rules
- Each team consists of three roles
  - Product owner (manage product backlog)
  - Development team
  - Scrum master

Scrum (2 of 4)

Scrum designed for speed and multiple functional product releases

Primary unit is the Sprint (runs 2 to 4 weeks, normally is 2 weeks)

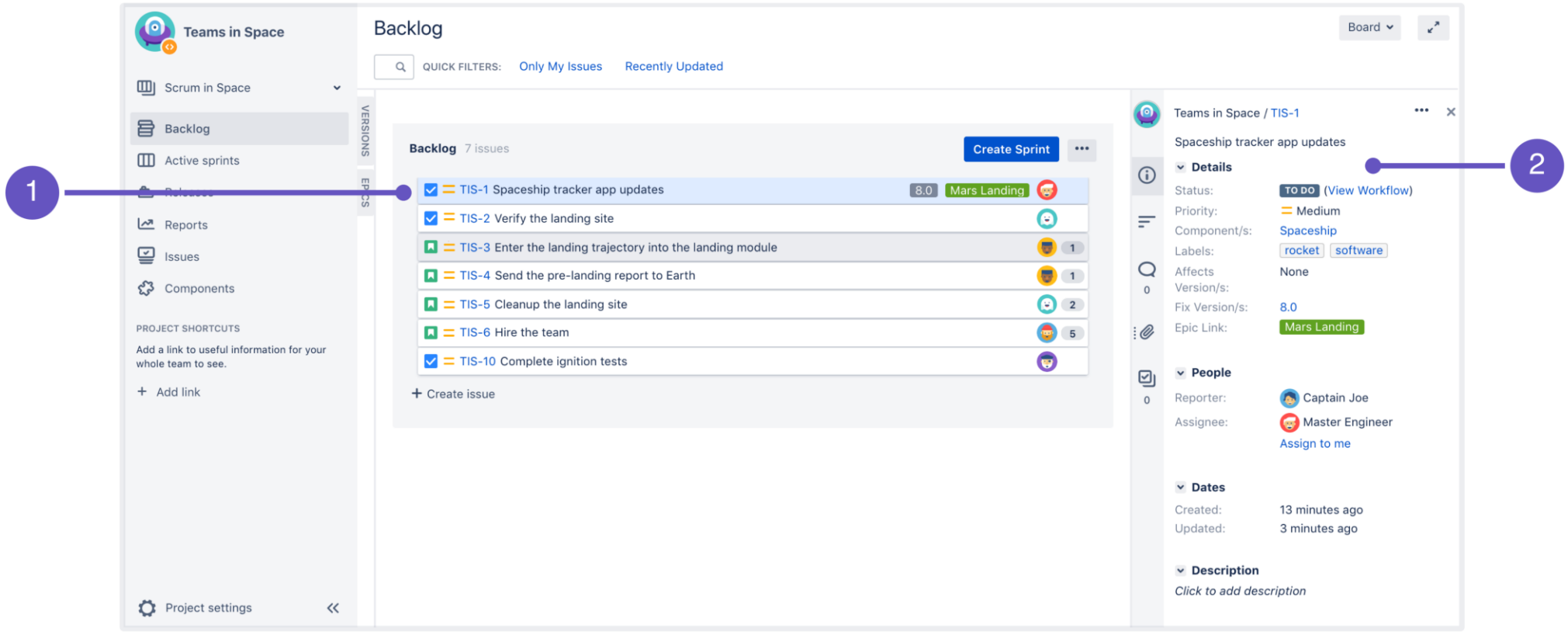
- Starts with an eight-hour planning meeting
  - What needs to be delivered by the end of the sprint
  - How will the team accomplish that work
- Daily Stand-up: A 15-minute meeting held to evaluate progress made within the past 24 hours and what needs to be done

Scrum (3 of 4)

- At the end of the sprint, two additional meetings
  - The Sprint Review: (4 hours) focusing on the product, what has been accomplished, and what needs to be done
  - The Sprint Retrospective: (3 hours) focusing on team performance and how it can improve
- Three primary artifacts in the Scrum process
  1. Product Backlog: Listing of potential requirements
  2. Sprint Backlog: Listing of only items to be addressed in a particular sprint
  3. Increment: Represents the sum of all the Product Backlog items completed during a sprint.

Scrum (4 of 4)

Daily Scrum Meeting
24 Hours
Product Backlog
Sprint Backlog
Potentially Shippable Product Increment
2-4 Weeks


Teams in Space
Scrum in Space
Backlog
Active sprints
Reports
Issues
Components
PROJECT SHORTCUTS
Add a link to useful information for your whole team to see.
Add link
Project settings
VERSIONS
EPICS
Backlog
QUICK FILTERS:
Only My Issues
Recently Updated
Board
Backlog 7 issues
Create Sprint
TIS-1 Spaceship tracker app updates
8.0
Mars Landing
TIS-2 Verify the landing site
TIS-3 Enter the landing trajectory into the landing module
TIS-4 Send the pre-landing report to Earth
TIS-5 Cleanup the landing site
TIS-6 Hire the team
TIS-10 Complete ignition tests
Create issue
Teams in Space / TIS-1
Spaceship tracker app updates
Details
Status:
TO DO
(View Workflow)
Priority:
Medium
Component/s:
Spaceship
Labels:
rocket
software
Affects Version/s:
None
Fix Version/s:
8.0
Epic Link:
Mars Landing
People
Reporter:
Captain Joe
Assignee:
Master Engineer
Assign to me
Dates
Created:
13 minutes ago
Updated:
3 minutes ago
Description
Click to add description
1
2

References

Agile Practice Guide (2017).

Systems Analysis and Design (9th)